

# Jiajie Wu CV

## Contact Information

Email: [j.wu24@imperial.ac.uk](mailto:j.wu24@imperial.ac.uk)

## Education

PhD in Mathematics, Imperial College London

From 2024.09

M.S. in Mathematics, New York University

2022.09 - 2024.05

B.A. in Mathematics, New York University (with high honors)

2018.09 - 2022.05

## Research Interests

Active Matter, Bio-fluid Dynamics, Cilia Dynamics, Collective Locomotion in Fluids, Delay Systems, Nonlinear Dynamics

## Research Projects

### **Research at the NYU Applied Math Lab**

**2019.11-2024.06**

Fluid-structure Interactions within Bird Flocks in Linear Formation Flights

Hydrodynamics of Finite-length Pipes at Intermediate  $Re$

The Water Clocks

### **Research at Imperial College London**

**Since 2024.10**

The Collective Dynamics of Cilia

### **Research at Helmholtz Pioneering Campus (Imperial-TUM Partnership)**

**Since 2024.11**

The Physics of Mucus

## Publications

C. Mavroyiakoumou, **J. Wu**, and L. Ristroph. "Modeling flying formations as flow-mediated matter," *Phys. Rev. Fluids* 11, 063103 (2026).

J. W. Newbolt, N. Lewis, M. Bleu, **J. Wu**, C. Mavroyiakoumou, S. Ramanarivo, and L. Ristroph. "Flow interactions lead to self-organized flight formations disrupted by self-amplifying waves," *Nat. Commun.* 15, 3462 (2024).

O. Pomerenk, S. Carrillo Segura, F. Cao, **J. Wu**, and L. Ristroph. "Hydrodynamics of finite-length pipes at Intermediate Reynolds Numbers," *J. of Fluid Mech.* (2023), vol. 959, A28.

## Awards and Honors

Imperial-TUM Joint Academy of Doctoral Studies Scholarship (MAAP\_G44167)

2024-2027

Roth Scholarship (MATH\_G00700)

2024-2028

Graduate School of Arts & Science Tuition Scholarship

2022-2023

Phi Beta Kappa Honors Society

2022.04

Founders Day Award (University Honors Scholar)

2022.04

Larry & Klara Silverstein Research Scholar

2020.05

Dean's Undergraduate Research Fund

2020.05

Dean's List for Academic Year (all 4 academic years, 2018 - 2022)

## Research Talks

**The Physics of Mucus**, Joint Academy of Doctoral Studies Summer School

2026.07

**The Water Clocks**, Dean's Undergraduate Research Conference

2021.05

## Work Experience

**Math & English Tutor, Shanghai Qingtu Education**

Summer 2019

**Course Grader, NYU Math Department**

MA-UY 1124 Calculus II for Engineers

Fall 2020

MA-UY 4414 Applied Partial Differential Equations

Fall 2020

**Graduate Employee Adjunct Faculty, Courant Institute of Mathematical Sciences**

|                                             |             |
|---------------------------------------------|-------------|
| MATH-UA 9 Algebra, Trigonometry & Functions | Fall 2022   |
| MATH-UA 264 Chaos and Dynamical Systems     | Spring 2023 |
| MATH-UA 132 Math for Economics II           | Fall 2023   |

**Graduate Teaching Assistant, Imperial College London**

|                                                             |             |
|-------------------------------------------------------------|-------------|
| MATH50004 Multivariable Calculus and Differential Equations | Fall 2024   |
| MATH40007 Introduction to Applied Mathematics               | Spring 2025 |
| MATH50003 Linear Algebra and Numerical Analysis             | Spring 2025 |
| MATH50003 Linear Algebra and Numerical Analysis             | Fall 2025   |
| MATH50003 Linear Algebra and Numerical Analysis             | Spring 2026 |

**References**

Dr. Eric Keaveny, Professor in Applied Mathematics, Imperial College London, [e.keaveny@imperial.ac.uk](mailto:e.keaveny@imperial.ac.uk)

Prof. Leif Ristroph, Associate Professor of Mathematics, New York University, [lr1090@nyu.edu](mailto:lr1090@nyu.edu)

Prof. Michael Shelley, Lilian and George Lyttle Professor of Applied Mathematics, New York University and Director of Center of Computational Biology, Flatiron Institute, [mshelley@flatironinstitute.org](mailto:mshelley@flatironinstitute.org)